Jack Peters

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Education

Boston University Boston, MA

B.S. in Mechanical Engineering, Concentration in Robotics — GPA: 3.25/4.00 *Expected May 2028*

- **Relevant Coursework:** Engineering Mechanics, CAD & Machine Components, Electric Circuits, Electromechanical Design, Introduction to Robotics, Manufacturing Processes, Fluid Mechanics, Energy & Thermodynamics

Technical Skills

CAD: SolidWorks, Fusion 360, Onshape |

Programming: C++, Python, MATLAB, Java, Assembly
Hardware: Arduino, ESP32, PCB Design, Embedded Systems, Sensor Integration, IMU/Motor Control
Manufacturing: CNC Milling, 3D Printing, Laser Cutting, Soldering

Experience

Regenity Biosciences Allendale, NJ

Equipment Engineering Intern *June 2026 – August 2026*

- Supported commissioning, qualification, and validation (CQV) of manufacturing equipment, ensuring reliable operation and compliance with internal quality standards
- Executed equipment testing, calibration, troubleshooting, and preventive maintenance, documenting results in formal validation protocols and technical reports
- Reviewed SOPs and calibration intervals for **469 equipment items**, supporting approvals that **doubled calibration frequencies** while maintaining full compliance

Indian Trail Club Franklin Lakes, NJ

Tennis Instructor *June 2022 – August 2026*

- Instructed 20+ beginner and intermediate students per season, adapting lessons to individual skill levels

Projects

Wireless Robotic Arm | *C++, ESP32-S3, MQTT, Computer Vision, Inverse Kinematics* September 2026 – October 2026

- Assembled and programmed a 3-DOF robotic arm using an ESP32-S3, servos, and a custom soldered Perfboard PCB
- Implemented MQTT communication to receive and parse joint commands, enabling wireless control of the base, shoulder, and elbow servos
- Developed camera calibration and AprilTag-based homography to transform image coordinates into the arm's world frame for automated motion
- Integrated inverse kinematics, vision data, and servo control to perform automated pick-and-place tasks using a magnetic end effector

3-Axis CNC Mill | *SolidWorks, Arduino Mega, Stepper Motors, DC Motor* September 2026 – October 2026

- Designed and built a 3-axis CNC mill with belt-driven linear slides and four stepper motors
- Developed SolidWorks mechanical designs and Arduino Mega control for coordinated motion and spindle operation
- Implemented Arduino Mega-based control of the motion system and spindle, integrating mechanical, electrical, and embedded systems into a functional CNC prototype

Electromechanical Desktop Game System *(Client-Based Design Project)* January 2026 – May 2026

- Designed a mechanically actuated 2-axis basketball hoop with motor-driven linear motion and a user-controlled catapult
- Integrated sensors and a microcontroller to automatically detect shots and update score/game state in real time
- Delivered a compact (12 in.) system to client specifications, combining mechanical actuation, embedded programming, and a user interface

Acrylic Truss Design and Buckling Analysis January 2026 – May 2026

- Conducted buckling tests and uncertainty analysis to characterize acrylic members and critical loads
- Designed and optimized a planar truss under geometric and material constraints to maximize load capacity

Leadership

Boston University Terrier Motorsport Boston, MA

Electrical and Grounded Low Voltage Member *September 2024 – September 2026*

- Evaluated and selected electrical components for an electric Formula SAE race car, supporting subsystem design decisions
- Prototyped and tested PCBs and electrical components, collaborating with a cross-functional team to refine designs
- Designed wiring harness layouts to improve electrical reliability, organization, and serviceability

Boston University Robotics Club Boston, MA

ARCTOS Mechanical Team Member *September 2026 – Present*

- Collaborate with the mechanical team on the development of a 6-DOF robotic arm
- Evaluate replacing 3D-printed forearm components with machined aluminum parts to improve structural strength and durability